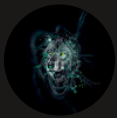




Build a Virtual Private Cloud (VPC)



Kiprono Yegon

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

nextwork-vpc

IPv4 CIDR block Info

☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block Info

☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

Tenancy Info

Default

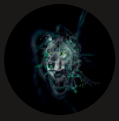
VPC encryption control (\$) Info

Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None ☐ Monitor mode ☐ Enforce mode

See which resources in your VPC are unencrypted but allow the creation of unencrypted resources.

Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.



Introducing Today's Project!

In this project, I will demonstrate setting up and managing a PC. I'm doing this project to learn cloud networking on AWS

What is Amazon VPC?

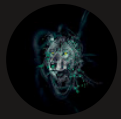
Amazon VPC is a virtual private network and it is useful because it is used to keep resources isolated

In today's project, I used Amazon VPC to create a private virtual network with subnets and internet gateways for internet connection

Personal reflection

This project took me less than 1 hour to complete

One thing I didn't expect in this project was the depth of the information on the VPCs and the Internet Gateways



Virtual Private Clouds (VPCs)

What I did in this step

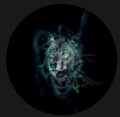
In this step, I will get into the VPC console in AWS because I will use the console to create the VPC

How VPCs work

VPCs are isolated virtual network hosted by a cloud provider for users to manage their resources and connectivity

Why there is a default VPC in AWS accounts

There was already a default VPC in my account ever since my AWS account was created. This is because some AWS services require the VPC for them to be functional; like the EC2 and some databases services, the VPC helps in the connection of services.



Create VPC [Info](#)

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VPC settings

Resources to create [Info](#)
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nextwork-vpc

IPv4 CIDR block [Info](#)
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10.0.0.0/16
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IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
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Tenancy [Info](#)
Default

VPC encryption control (\$) [Info](#)
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#) [L2](#)

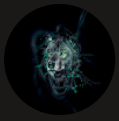
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See which resources in your VPC are unencrypted but allow the creation of unencrypted resources.

Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

Defining IPv4 CIDR blocks

To set up my VPC, I had to define an IPv4 CIDR block, which is a way to assign a whole block of IP addresses.



Subnets

What I did in this step

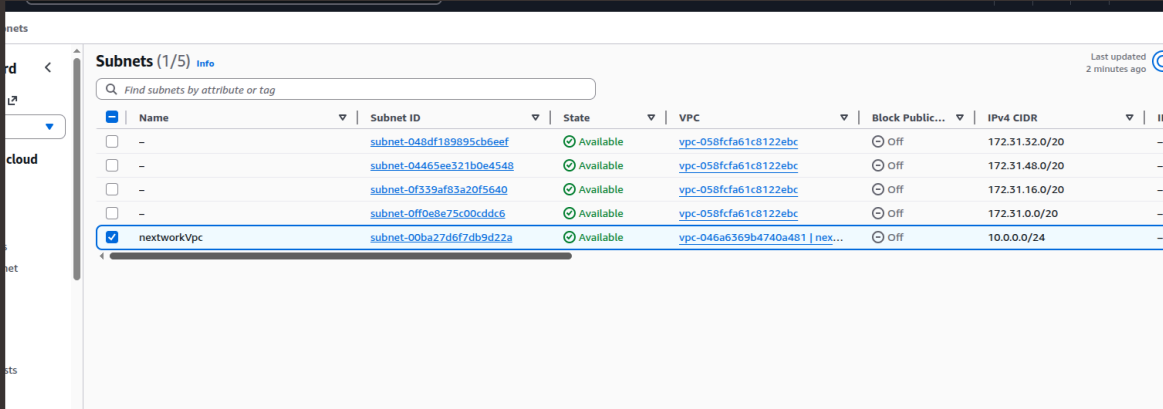
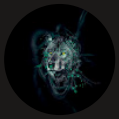
In this step, I will launch a subnet in the VPC because subnets are used to group resources with similar access rules and restrictions.

Creating and configuring subnets

Subnets are groupings of resources that have similar access rules and restrictions. There are already subnets existing in my account, one for every Availability Zone.

Public vs private subnets

The difference between public and private subnets are resources inside the public subnet can communicate with external resources whereas for private subnet the resources are not to be accessed publicly. For a subnet to be considered public, it has to have access to the internet via the internet gateway

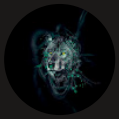


The screenshot shows the AWS Subnets console. The title is 'Subnets (1/5)'. There is a search bar with the placeholder text 'Find subnets by attribute or tag'. The table has columns: Name, Subnet ID, State, VPC, Block Public..., IPv4 CIDR, and IP. There are five rows of subnets. The last row, 'nextworkVpc', is selected with a blue checkmark in the first column.

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IP
☐	-	subnet-048df189895cb6eef	Available	vpc-058fca61c8122ebc	☐ Off	172.31.32.0/20	-
☐	-	subnet-04465ee321b0e4548	Available	vpc-058fca61c8122ebc	☐ Off	172.31.48.0/20	-
☐	-	subnet-0f339af83a20f5640	Available	vpc-058fca61c8122ebc	☐ Off	172.31.16.0/20	-
☐	-	subnet-0ff0e8e75c00cddc6	Available	vpc-058fca61c8122ebc	☐ Off	172.31.0.0/20	-
☒	nextworkVpc	subnet-00ba27d6f7db9d22a	Available	vpc-046a6369b4740a481 nex...	☐ Off	10.0.0.0/24	-

Auto-assigning public IPv4 addresses

Once I created my subnet, I enabled auto-assign public IPv4 address. This setting makes sure that any EC2 instance launched in that subnet will instantly get a public IP address so that it is accessible via the internet and you don't have to create one manually.



Internet gateways

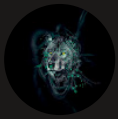
What I did in this step

In this step, I will connect my VPC to the internet using an internet gateway because the internet gateway will make the applications in the VPC available on the internet.

Setting up internet gateways

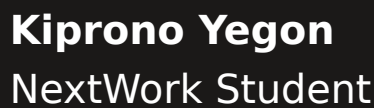
Internet gateways are key to making applications available on the internet. By attaching an internet gateway, your instances can access the internet and be accessible to external users.

Attaching an internet gateway to a VPC means that my VPC is now connected to the internet. If I missed this step then my VPC will not be connected to the internet and cannot be accessed over the internet



Internet gateways (1/2) [Info](#)

☐	Name	Internet gateway ID	State	VPC ID	Owner
☐	-	igw-0aede5adb496b3160	Attached	vpc-058fca61c8122ebc	783259868137
☒	nextworkIGW	igw-0d118e970a1a4383a	Attached	vpc-046a6369b4740a481 nextwork-vpc	783259868137



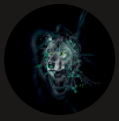
What I'm doing in this extension

In this project extension, I will use the AWS CLI (AWS CloudShell) to launch VPC's resources

Exploring CloudShell and CLI

VPC resources could also be created with CloudShell, which is a space to run code which has AWS CLI pre-installed. CLI is a command line interface that lets you create, delete and update AWS resources with commands instead of clicking through your console.

```
aws ec2 create-vpc --cidr-block 10.0.0.0/24 --query Vpc.VpcId --output text
aws ec2 create-subnet --cidr-block 10.0.0.0/16 --query Subnet.SubnetId
aws: [ERROR]: An error occurred (InvalidID) when calling the CreateTags operation: The ID 'VPC-ID' is not valid
aws ec2 create-tags --resources VPC-ID --tags Key=Name,Value=NextWork VPC 2
aws: [ERROR]: An error occurred (InvalidID) when calling the CreateTags operation: The ID 'VPC-ID' is not valid
aws ec2 create-tag --resources VPC-ID --tags Key=Name,Value=NextWork VPC 2
aws: [ERROR]: An error occurred (InvalidID) when calling the CreateTags operation: The ID 'VPC-ID' is not valid
aws ec2 create-subnet --vpc-id VPC-ID --cidr-block CIDR-CIDR-BLOCK-HERE
aws: [ERROR]: An error occurred (InvalidParameterValue) when calling the CreateSubnet operation: Value (CIDR-CIDR-BLOCK-HERE) for parameter cidrBlock is invalid. This is not a valid CIDR block.
aws ec2 create-subnet --vpc-id VPC-ID --cidr-block CIDR-CIDR-BLOCK-HERE
aws: [ERROR]: An error occurred (InvalidSubnetRange) when calling the CreateSubnet operation: The CIDR '10.0.0.0/16' is invalid.
aws ec2 create-subnet --vpc-id VPC-ID --cidr-block CIDR-CIDR-BLOCK-HERE
{
  "Subnet": {
    "AvailabilityZoneId": "usw2-az1",
    "MapCustomerOwnedIpv6Prefix": false,
    "OwnerId": "78325986137",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6Prefixes": [],
    "SubnetArn": "arn:aws:ec2:us-west-2:78325986137:subnet-09ef40b940ba783f5",
    "EnableDns": false,
    "Ipv6Native": false,
    "PrivateDnsHostnames": {
      "Hostnames": {
        "Type": "ip-name",
        "EnableResourceNameMappings": false,
        "EnableResourceNameMappings": false
      }
    },
    "SubnetId": "subnet-09ef40b940ba783f5",
    "State": "available",
    "VpcId": "vpc-0023e584b408612c8",
    "CidrBlock": "10.0.0.0/24",
    "AvailableIpAddressCount": 251,
    "AvailabilityZone": "us-west-2b",
    "DefaultForAz": false,
    "MapPublicIpOnLaunch": false
  }
}
aws ec2 create-internet-gateway --query InternetGateway.InternetGatewayId --output text
aws ec2 attach-internet-gateway --vpc-id VPC-ID --internet-gateway-id IG-ID
aws: [ERROR]: An error occurred (InvalidInternetGatewayId,Malformed) when calling the AttachInternetGateway operation: Invalid id: "IG-ID" (expecting "Igw-...")
aws ec2 attach-internet-gateway --vpc-id VPC-ID --internet-gateway-id IG-ID
aws ec2 attach-internet-gateway --vpc-id VPC-ID --internet-gateway-id IG-ID
aws: [ERROR]: An error occurred (ResourceAlreadyAssociated) when calling the AttachInternetGateway operation: resource Igw-02f9e7490a7764420 is already attached to network Vpc-0023e584b408612c8
aws ec2 create-internet-gateway --query InternetGateway.InternetGatewayId --output text
aws ec2 attach-internet-gateway --vpc-id VPC-ID --internet-gateway-id IG-ID
aws: [ERROR]: An error occurred (InvalidInternetGatewayId,Malformed) when calling the AttachInternetGateway operation: Invalid id: "IG-ID" (expecting "Igw-...")
aws ec2 attach-internet-gateway --vpc-id VPC-ID --internet-gateway-id IG-ID
aws: [ERROR]: An error occurred (ResourceAlreadyAssociated) when calling the AttachInternetGateway operation: resource Igw-02f9e7490a7764420 is already attached to network Vpc-0023e584b408612c8
```



Comparing CloudShell vs AWS Console

Compared to using the AWS Console, an advantage of using commands is the CLI can be used for writing scripts. An advantage of using the Console is it has AWS CLI pre-installed Overall, I preferred using the AWS Cloudshell



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